

Name: M. Ali Amir

Date: 23-1-2026

Subject: AI

Quiz # ~~100~~

Q:

5

Given that

$$K = 3$$

- No of Purchases
- Time spent (hours)

New Customer F

- Purchases = 4
- Time spent = 3

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Customer	Distance	Category
A	$\sqrt{(4-2)^2 + (3-1)^2} = \sqrt{8}$ $= 2.83$	L
B	$\sqrt{(4-3)^2 + (3-2)^2} = \sqrt{2}$ $= 1.41$	L
C	$\sqrt{(4-6)^2 + (3-5)^2} = \sqrt{8}$ $= 2.83$	H
D	$\sqrt{(4-7)^2 + (3-6)^2} = \sqrt{18}$ $= 4.24$	H
E	$\sqrt{(4-5)^2 + (3-4)^2} = \sqrt{2}$ $= 1.41$	H

$$\therefore K = 3$$

Nearest 3 distances:

① $B(1.41) \quad L$

② $E(1.41) \quad H$

③ $A(2.83) \quad L \quad (\text{tie broken arbitrarily})$

Majority vote

$$\text{Low}(L) = 2$$

$$\text{High}(H) = 1$$

Customer F is classified as Low Value (L).